cetelem

Hello!

Today’s challenge: New ways of Reducing Churn

What is Churn?

Customer churn **rate** (or attrition):

How many **customers**
Stop doing business
with a company or
Stop using a service
over a given period.

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Why does it matter?

Revenue Loss	Acquisition Cost	Market Share
Directly impacts profitability Less clients or Less Services	**Retention** typically is **significantly cheaper** than acquiring new clients	If the need still exists, we are probably **losing clients** to competitors
Decreased revenue	**Increasing Costs**	**Loss of Share**

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Why understand churn and model it?

Allows to try to change a behavior, before it happens!

Proactive

Identify at-risk customers

Who will churn?

When will they churn?

Create models

Targeted

Implement retention strategies

Who is churning?
Why are they churning?
What can I do?
How do they react?

Analytics and model explanation

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Types of Churn

Involuntary Churn:

Customers leave due to circumstances beyond their control (e.g., failed payments, relocation, service termination).

Voluntary Churn:

Customers actively decide to leave (e.g., cancel an insurance, stop using a card).

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What to Model ?

If question:

“Is this customer likely/going to churn?”

Traditional models to get a probability and binary classification.

Needs to have a fixed predefined outcome “in the next 60 days”

When question:

“When is this customer likely to churn?”

Survival Analysis Models – (Kaplan-Meier Estimator, Cox Prop. Hazards, Aalen’s Additive)

They analyze the time until an event occurs, providing a dynamic view of churn risk

Not to Forget! – We want to prevent churn before it happens,
before the decision to churn is taken.

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In Practice:
The Business Case!

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Data and Context

Churn in Portugal

Target:

Traditional Loans

Objectives:

Find and understand churners

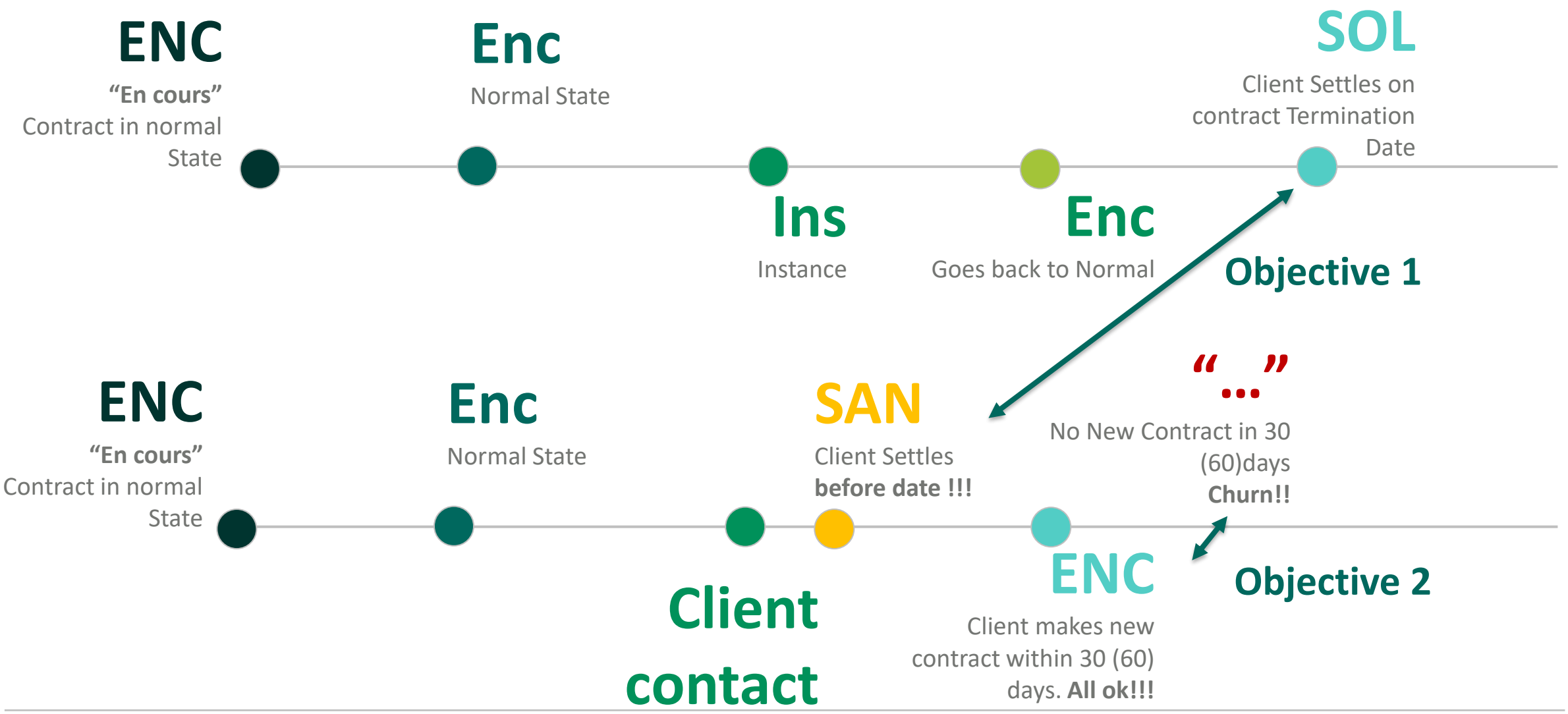
Data From:

Portugal, last years

If Pilot goes well:

Expand to other major countries

The Business Case



Objectives

Objective 1

Can you Identify groups in clients that settle earlier?
How are they different from groups that go to the end?

Objective 2

Can you Identify groups in clients that go for a new contract, and the ones that do not? (Classification or Clustering)

How are they different and what are your recommendations to approach each group?

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The Data

BDOSS

Information on each Contract

One Line corresponds to one observation of a contract on the last day of that month

Information on:

150K clientes

185K contracts

23 Months: 01-2024 to 11-2025

Total: 2.5M+ rows

Column “POS” : contract state

CRC

Information from BDP

One Line Corresponds to one observation from Bank of Portugal on the registered credits outstanding.

Information on:

140K Clients

25 Months: 12-2023 to 12-2025

Total: 3M+ rows

CScore

Client risk scores

One Line corresponds to one customer at the date of last credit proposal.

Information on:

62K Clients

1 Month: ~Last Proposal(s)

Total: 63K rows

Fama

Client Financial aggregated values

One Line corresponds to one customer at the end of 2024.

Information on:

141K Clients

1 Month: 11-2024

Total: 141K rows

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Data Limitations

We had to manipulate some data in order to protect:

Our Customers and their rights

The bank **business information**

In order to provide you with recent, quality data:

- Some values were manipulated, including wages, amounts, ages in order to retain order and useful information.
- Some variables meanings were obfuscated and values hashed/mapped
- This is a selected sample of our clients and contracts

You may treat it as if the data was original, complete and valid.
We can from your results extrapolate to our client universe.

NOTE: It still remains confidential, and you are bound by your NDA.

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Thank you!